

Combining Covariate Adjustment with Information-Adaptive Designs

RISW, September 2026

Planning

- 1 Background, Problem Setting and Set Up
- 2 Potential Solution: Information-Adaptive Design
- 3 Simulation Study

Outline

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Covariate Adjustment

- **Covariate adjustment** is a statistical analysis method with high potential to **improve precision** for many trials.
 - **Pre-planned** adjustment for baseline variables when estimating **average treatment effect**.
 - Estimand is same as when using unadjusted estimator (e.g., difference in means).
 - **Goal**: avoid making any model assumptions beyond what's assumed for unadjusted estimator (**robustness to model misspecification**).

(e.g., Koch et al., 1998; Yang and Tsiatis, 2001; Rubin and van der Laan, 2008; Tsiatis et al., 2008; Moore and van der Laan, 2009b,a; Zhang, 2015; Jiang et al., 2018; Benkeser et al., 2020)

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- Estimator: G-computation/Standardization

- 1 Fit logistic regression model for

$$P(Y = 1|A, X) = \text{logit}^{-1}(\gamma_0 + \gamma_1 A + \gamma_2 X).$$

- 2 Compute standardized estimators for treatment specific means

- $\hat{E}(Y|A = 1) = \frac{1}{n} \sum_{i=1}^n \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_1 + \hat{\gamma}_2 X_i)$

- $\hat{E}(Y|A = 0) = \frac{1}{n} \sum_{i=1}^n \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_2 X_i)$

- 3 Calculate $\hat{\theta} = \hat{E}(Y|A = 1) - \hat{E}(Y|A = 0)$

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 - **Approach 1:** assume conservatively that covariate adjustment will not lead to a precision gain.
 - **Approach 2:** consider how much precision can be gained based on external (trial) data when calculating the sample size. (Li et al., 2023)
 - An incorrect projection of a covariate's prognostic value, may still lead to an over- or underpowered future trial.

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 - **Approach 1:** assume conservatively that covariate adjustment will not lead to a precision gain.
 - **Approach 2:** consider how much precision can be gained based on external (trial) data when calculating the sample size. (Li et al., 2023)
 - An incorrect projection of a covariate's prognostic value, may still lead to an over- or underpowered future trial.
 - **Potential solution:** combine covariate adjustment with **information-adaptive designs** (also known as information monitoring).

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Algorithm for Analysis Timing: Design Stage

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- We **compute the maximum/total information** needed to preserve these operational characteristics

$$\left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 ,$$

for a fixed design (no interim analyses), and

$$\left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 IF$$

when data is sequentially monitored with the possibility of early stopping.

(Mehta and Tsiatis, 2001)

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- We conduct the final analysis at time t_2 when

$$(\widehat{se}(\hat{\theta}_{t_2}))^{-2} \geq \left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 IF.$$

Algorithm for Analysis Timing: (Dis)advantages

- The **information-adaptive design** is well suited for being adopted for covariate adjusted estimators:
 - We do **not have to prespecify the prognostic value of the covariates** nor other nuisance parameters.
 - When the estimator is more efficient than unadjusted estimator, covariate adjustment can lead to a **shorter trial** due to faster information accrual.

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- The **information-adaptive design** is well suited for being adopted for covariate adjusted estimators:
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 - When the estimator is more efficient than unadjusted estimator, covariate adjustment can lead to a **shorter trial** due to faster information accrual.
- **Administrative inconvenience**: it does not give an idea to the investigators about the necessary resources (i.e., length of study, sample size, ...).

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 - We recommend setting the **sample size conservatively** as if there were no precision gain from covariate adjustment.
- However, **miscalculations can occur** at the design stage.
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- We should use the **emerging data to evaluate** whether the maximum information will be reached with the planned sample size.
- If not, we should **update the maximum sample size** at time t as

$$n_{max} = n(t) \frac{\left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 IF}{(\widehat{se}(\hat{\theta}_t))^{-2}},$$

where $n(t)$ is the number of patients used in the analysis at time t .

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- We proposed an **information-adaptive trial design**, where the analysis timing is based on **accruing information** and is **data-adaptive**.
- By **automatically adapting to amount of precision gain** due to covariate adjustment, it results in **correctly powered** trials.
- Information will accrue faster as covariate adjusted estimators typically have smaller variance, leading to **faster trials at no additional cost**.
- Since adaptations to the analysis timing are **pre-planned** based on nuisance parameters only, they are generally **acceptable to regulators**.

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MISTIE III trial (Stroke)

- Functional outcome: proportion of patients who achieved a modified Rankin Scale score of 0-3 at 365 days (**binary**).
- Estimand of interest: **risk difference**.
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 - Success rate: 25% in standard medical care group versus 38% in MISTIE group
- We will focus on information instead of sample size!

Simulation Study: $K = 1$

- **Information-adaptive design** with maximum information equal to 582
- **Maximum sample size design** with $n_{max} = 498$

		$\theta = 0.13$ (Alternative)			
		Power	ASN	AAT	AI
Information-adaptive design	Unadjusted	88.4%	571	1876	582
	Standardization	87.3%	433	1509	567
Maximum sample size design	Unadjusted	83.1%	-	1682	508
	Standardization	91.1%	-	1682	652

ASN: average sample number; AAT: average analysis time (days); AI: average information.

Conclusion under alternative:

24% reduction of sample size due to covariate adjustment

Simulation Study

- **Information-adaptive design** with maximum information equal to 582
- **Maximum sample size design** with $n_{max} = 498$

		$\theta = 0$ (Null)			
		Type I	ASN	AAT	AI
Information-adaptive design	Unadjusted	5.28%	569	1871	582
	Standardization	5.28%	402	1427	568
Maximum sample size design	Unadjusted	5.14%	-	1682	509
	Standardization	5.14%	-	1682	705

ASN: average sample number; AAT: average analysis time (days); AI: average information.

Conclusion under null:

29% reduction of sample size due to covariate adjustment

Thank you for your attention!

Interested? Van Lancker, K., Betz, J. F., & Rosenblum, M. (2025). Combining covariate adjustment with group sequential, information-adaptive designs to improve randomized trial efficiency. Biometrics.

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